Rooh ਰੂਹ

Outlines root cross-sections automatically, lets you fix them by hand, and gives you the traits as Excel. Use a laptop or desktop in Chrome or Firefox (you need a keyboard and mouse).

`jagdeep2021--rooh.modal.run` the website; the desktop app is the same page

STEPS

1. **Ask for access (once)**

   Type your name and email, click `Ask for access`. Jagdeep approves it, then you reload the page. (Not needed in the desktop app.) Keep using the *same browser*: it remembers you. A private window or cleared cookies means asking again.

2. **Make a project**

   Click `+` at the top, type a name (e.g. the experiment ID), `Create`.

3. **Add images**

   `Folder...` uploads a whole folder; each sub-folder becomes a group (genotype, treatment) in the results. `Files...` adds single images. TIF, PNG or JPG. Don't close the tab until the upload finishes.

4. **Outline**

   Uploading does **not** start this: when the upload finishes, click `Outline`. The server finds cells, stele, vessels and root edge, about 10 s per image. If you add more images later, click it again; it only does the new ones and leaves your edits alone.

5. **Set the scale**

   Type the µm per pixel in the box, or click `📏` (or `M`), drag along the scale bar and type its length. Then `All` for every image, or `Image` for this one only. Without a scale, areas come out in pixels.

6. **Check every image**

   Pick what to fix in the Edit panel (or press its number), then: **drag** draws, **click** adds, hold `⌘`/`Ctrl` to remove.

   - `1` Cell: click inside to fill
   - `2` Vessel (metaxylem)
   - `3` Stele: draw the outline
   - `4` Sector: leave out damage
   - `5` Root: draw the outline
   - `6` Aerenchyma lacunae

Get the **stele right first**: cell file numbers and positions are counted out from it. When the image looks right, `Save + next` (`Shift`+`S`). In the image list ● = checked, ○ = not yet.

7 **Make results, then Download**

`Make results` builds the Excel files and group graphs; `Download` gives you a zip. Re-run it after any more edits; never edit the Excel by hand.

RULES THAT MATTER FOR THE NUMBERS

- **Leave out technical damage only** (torn, folded, oblique, out of focus, debris). Aerenchyma, collapsed or odd cells are biology: keep them, they get measured.
- To leave out damage: Sector tool, drag around the stele from one side of the damage to the other, and pick the reason. `⌘`/`Ctrl`-click a grey arc to bring it back.
- Aerenchyma: orange = counted. If an orange patch is really cells the outlining missed, fix it with the Cell tool rather than un-counting it.
- Don't delete a project ( ) without asking Jagdeep.

MOVING AROUND

`Space`-drag or right-drag	move the image
scroll	zoom
`F`	fit the image to the window
`0`	show / hide outlines
`,` `.`	previous / next image (also saves edits)
`Shift`+`A` · `Shift`+`F`	aerenchyma layer · colour cells by file number

Something wrong? Send Jagdeep the project name, the image name and a screenshot.